



AI/NLP Financial Risk Engine

From news headlines and tweets to index weights and stress losses

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S&P Global & Crisil Campus Hackathon · Case study: real-time risk signals

Markets move on text, but risk systems need numbers

The problem

- Thousands of headlines and posts a day; a risk desk can read a handful.
- Text is unstructured: no field says how bad, what kind, or which asset.
- By the time a human has read it, prices have moved.

My approach

- One engine turns every document into a signal: sentiment (-1...+1), event type, impact (1-10).
- Two real sources: 20k news headlines + 60k tweets, 2014-2016, replayable as a live stream.
- Built both modules: an index rebalancer (A) and a banking-book stress tester (B).

Input

"Brazil downgraded to junk status by 2nd risk agency, with negative outlook for further downgrades."

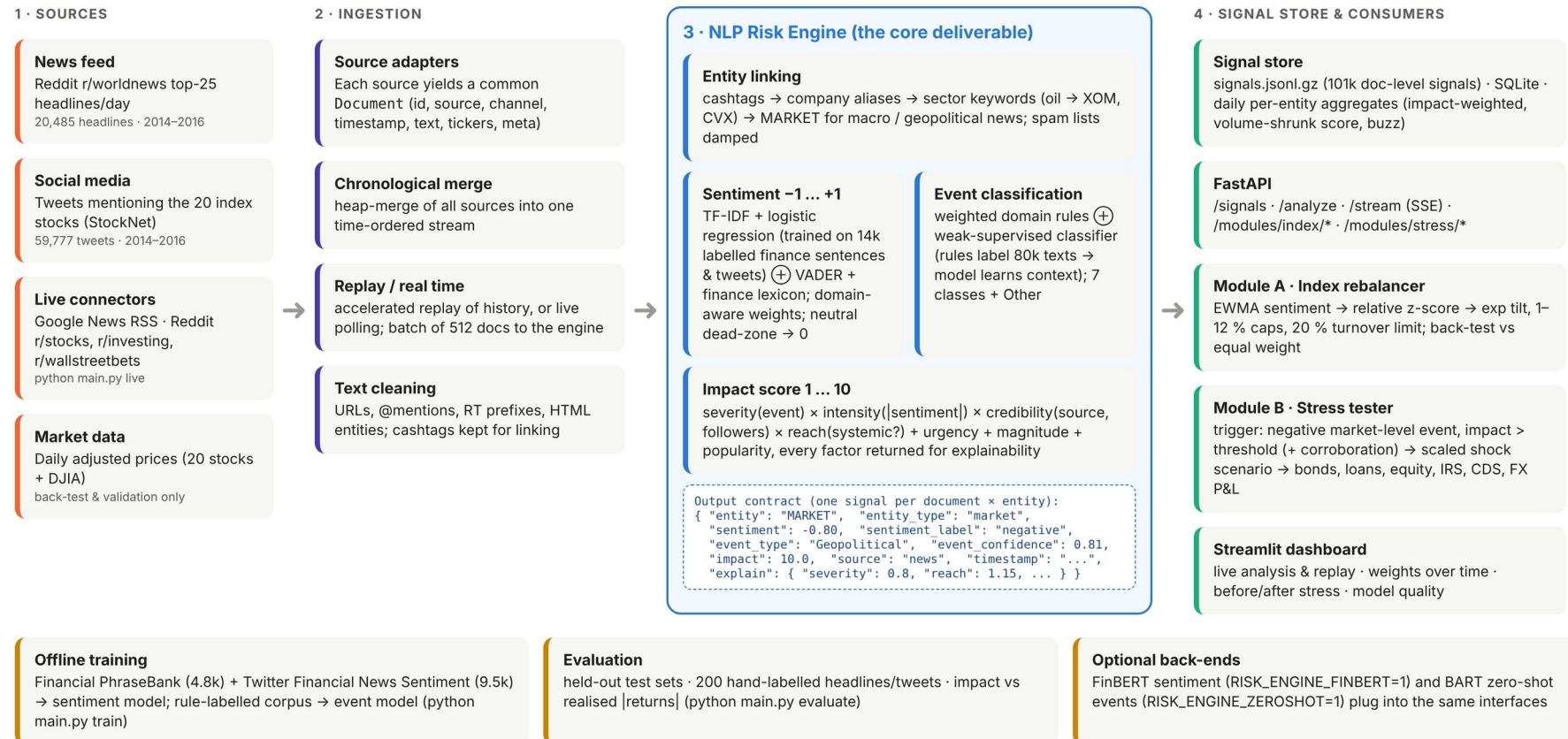
Engine output

Entity	MARKET (market-wide)
Sentiment	-0.96 negative
Event	Credit Event
Impact	10.0 / 10
Action	Stress test fires → -\$466m

System design: one engine, a signal contract, two consumers

AI/NLP Financial Risk Engine: architecture and data flow

Unstructured text in, structured risk signals out, consumed by two downstream modules. Everything runs locally in Python.



All data is public (StockNet MIT, Reddit WorldNews CC BY-NC-SA, Financial PhraseBank CC BY-NC-SA) or synthetic (banking portfolio). No proprietary or client data.

Implementation highlights and why I chose them

Sentiment

TF-IDF + logistic regression trained on 14k labelled finance sentences and tweets, blended with a finance-tuned VADER lexicon.

Why: runs on a laptop in ms, explainable, and the lexicon covers geopolitical text the model never saw. FinBERT plugs in as an option.

Event type

Weighted domain rules + a weak-supervised classifier: rules label 80k unlabelled texts, the model learns the context around them.

Why: no labelled event dataset exists for this period; weak supervision gets coverage without hand-labelling.

Impact 1–10

severity $\times$ intensity $\times$ credibility $\times$ systemic reach + urgency + magnitude + popularity. Every factor is returned.

Why: risk governance needs to explain a score; each weight is a domain assumption you can challenge.

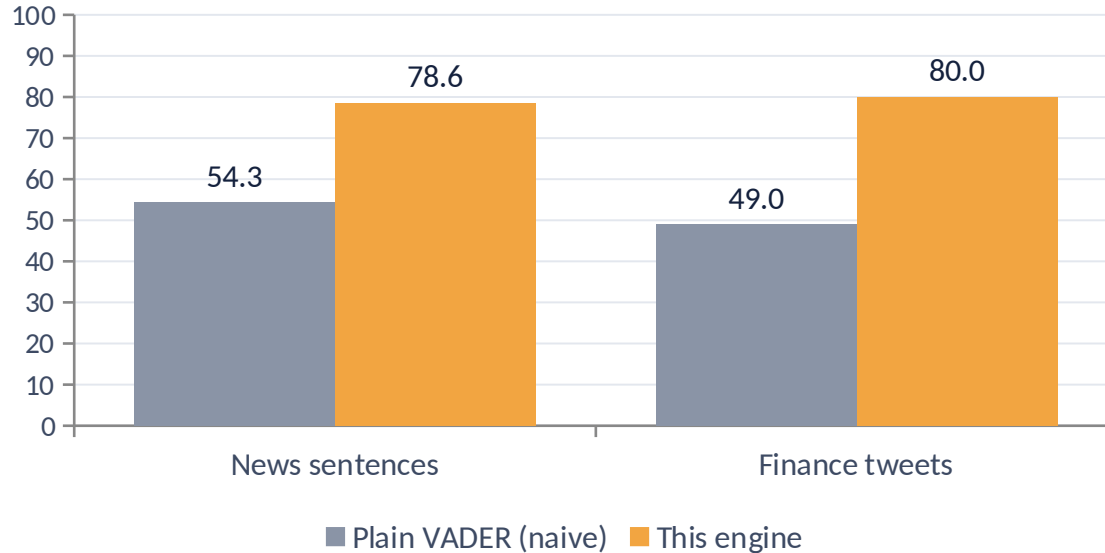
Modules

A: EWMA $\rightarrow$ relative z-score $\rightarrow$ exp tilt, 1–12 % caps, 20 % turnover limit. B: 203-trade book, shock library scaled by impact, sensitivity revaluation.

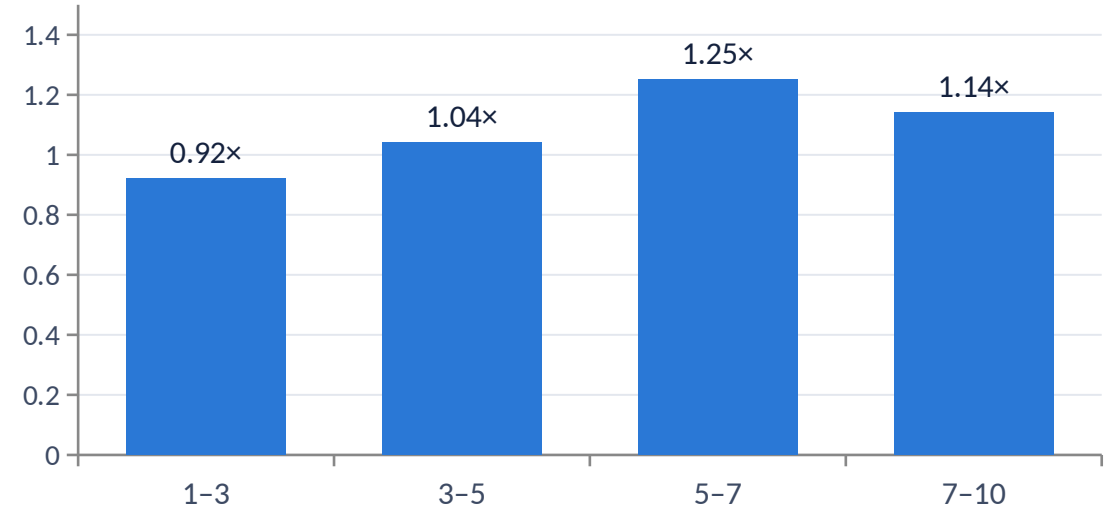
Why: standard, auditable portfolio and stress-testing mechanics; FastAPI + Streamlit for demo and integration.

Results: beats the naive baseline, checked against prices

Sentiment accuracy on held-out labelled data (%)



Same-day |return| by impact score, vs each stock's typical day



101,437

structured signals from 60,388 documents (~1 ms each)

67.5%

event accuracy on 200 hand-labelled texts

+0.41% / -0.37%

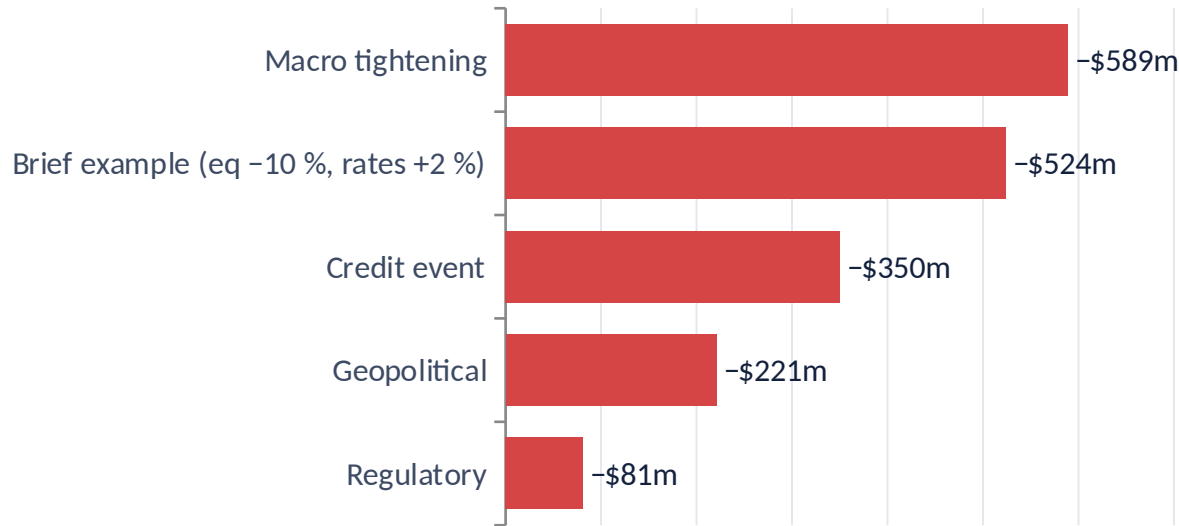
avg same-day return on positive / negative sentiment days

22.2% / 22.3%

Module A / equal-weight return (23.5% before costs)

Domain impact: from a headline to capital at risk

Loss on the \$8.1bn synthetic banking book (impact 8.5)



- 193 stress triggers on 178 days, including Russia-Ukraine (Mar 2014), the OPEC price war (Nov 2014), the Greek debt crisis and Brazil's junk downgrade (2015).
- Losses arrive broken down by risk factor, asset class, sector and trade, in the format risk committees already use.
- Every score explains itself, which model validation requires.

Analyse any headline or post

Text: Russia invades Crimea as Ukraine crisis escalates, markets plunge

Example: Russia invades Crimea as Ukraine crisis escalates

Source type: ☒ news ☐ social

entity	entity type	sentiment	label	event	event conf.	impact (1-10)	stress trigger?
MARKET	market	-0.7979	negative	Geopolitical	0.81	10	yes

> Why this score? (explainability)

1 • A headline is scored live in the dashboard



2 • The trigger revalues the book: before vs after, P&L by risk factor

Limitations and next steps

Out-of-domain sentiment

The supervised model reads geopolitical text poorly (33 %); routing to the lexicon lifts it to 48 %.



Fine-tune FinBERT on labelled world-news headlines.

Noisy news feed

Reddit world-news is not market-focused; market-level impact does not predict DJIA moves.



Add Reuters/GDELT feeds; learn impact weights from realised moves.

Signal $\neq$ forecast

Sentiment explains same-day returns but not the next day's (Spearman -0.02).



Use as a risk overlay; test intraday timing and cost-aware rebalancing.

Simplified stress model

Linear sensitivities, 1-year PDs, a synthetic book, a 200-text evaluation set.



Full revaluation, scenario correlation, larger multi-rater gold set.